Review on Evapotranspiration estimation by Deep Learning

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***Abstract*—This study investigates deep learning techniques employed to estimate the Evapotranspiration, the process by which water is transferred from the soil and plants to the atmosphere through evaporation or transpiration. Accurate estimation of evapotranspiration is crucial in agriculture, where it can help to determine optimal irrigation levels and reduce water waste. The research reviews commonly used prediction algorithms, performance metrics, and details about datasets and study areas. Moreover, a personal implementation of a Long Short-Term Memory (LSTM) model was developed and tested to perform predictions of the actual evapotranspiration in the Piedmont region in Italy.**

I. Introduction

AGRICULTURE is a critical sector with specific optimization needs to enhance crop yields and minimize water consumption. Evapotranspiration (ET) is a key component of the water cycle, it quantifies the volume of water vapor transferred from the Earth's surface to the atmosphere through evaporation and transpiration in a specific period of time [1]. Accurate estimation of ET is therefore essential for improving irrigation scheduling and helping in the early detection of drought conditions [2]. Additionally, in climate science and meteorology, ET plays a critical role in enhancing weather prediction models and monitoring the impacts of climate change [3].

However, directly measuring ET can be challenging because it necessitates the application of specific models that integrate various climatological and soil parameters.

Traditional ET estimation are primarily based on physical principles, such as the FAO-56 Penman-Monteith model, which integrates physiological characteristics to the FAO-24 Penman.

However, these methods frequently fail to capture the nonlinear relationships of the meteorological data and have significant limitations on performing ET estimations in absence of some of these data. For this reason, deep learning algorithms can be employed due to its ability to handle non-linear relationships and spot complex patterns [1].

II. Background

There are several different types of Evapotranspiration, each of these needs a different model to be estimated:

A. Reference Evapotranspiration (ET_0)

Represents the evapotranspiration from a standardized reference surface, like well-watered grass, and it serves as a benchmark for estimating crop water requirements.

The most widely used technique to evaluate ET_0 is the Penman-Monteith equation (FAO-56). This method depends on meteorological variables, as air temperature, relative humidity, solar radiation, and wind speed. Moreover, it does not require any local calibration because it incorporates both physiological and aerodynamic parameters [4].

However, these climatological data are not always available in all the study areas. Therefore, alternative approaches are available and can be classified in three types, namely, radiation-based models (Priestley–Taylor), temperature-based models (Hargreaves and Samani, Trajkovic), and combination models (Penman) [5] [6] [7].

B. Potential Evapotranspiration (PET)

Measures the maximum evapotranspiration possible from a surface, assuming unlimited water availability.

To evaluate PET, the same models used for ET_0 can be applied, but with surface parameters adapted to the actual vegetation and land cover.

C. Actual Evapotranspiration (AET)

Describes the real amount of water lost from a surface to the atmosphere, considering the limited water availability in the soil. Since it depends on water availability, AET is more complex to estimate and measuring it over large areas can be difficult and costly.

Lysimeters, which are sensors that accurately quantify water loss by tracking variations in the water volume contained in the soil, can be used to directly measure AET in a specific location. Although this method is precise and reliable, it is also quite expensive [8].

Another direct measurement technique is the Eddy Covariance (EC) method, which captures water vapor fluxes above the canopy through high frequency measurements. While EC provides detailed insight into evapotranspiration dynamics, it requires sophisticated instrumentation and advanced data processing [9].

Other solutions, for broader spatial coverage and with less complexity, are based on satellite remote sensing [10]. Satellites collect various spectral indices that provides insights into vegetation condition, surface temperature, and energy balance components–such as NDVI, EVI, LST, LAI, SPI, and SPEI (see table I) [11]. These indices can be used as input to specific models that can measure the AET, such as the Surface Energy Balance (SEB) model [12].

Another commonly used approach is the crop coefficient method, which estimates AET by adjusting reference evapotranspiration ET_0 with coefficients that account for the crop type, growth stage, and water stress conditions.

TABLE I
DESCRIPTION OF SPECTRAL AND CLIMATIC INDICES USED FOR AET ESTIMATION

Acronym	Full Name
NDVI	Normalized Difference Vegetation Index
EVI	Enhanced Vegetation Index
LST	Land Surface Temperature
LAI	Leaf Area Index
SPI	Standardized Precipitation Index
SPEI	Standardized Precipitation Evapotranspiration Index

III. Machine Learning and Neural Network Approaches for Evapotranspiration Estimation

Machine learning (ML) and neural networks (NN) are powerful tools for ET estimations due to their ability to handle complex, non-linear relationships in the data. Traditional ET estimation models often rely on meteorological parameters, that might be unavailable in some areas and can be costly to be measured. ML and NN approaches can address these limitations by performing pattern recognition, and accurate predictions even with incomplete data [13].

A. Traditional ML

1) Random Forrests (RF): Widely utilized for regression tasks, RF constructs an ensemble of decision trees, each trained on a randomly sampled subset extracted from the training set. To compose the final prediction, the outputs of all individual trees are aggregated, typically by averaging the results. RF models are known for their strong generalization capabilities, robustness to outliers, and ease of hyperparameter optimization, making them effective for a wide range of prediction tasks [14].

2) Support Vector Machine (SVM): Transforms the input vector into an higher-dimensional feature space, and finds the optimal hyperplane that separate these points. Generally, SVM can be mathematically explained with the Eq. 1

$$Y = w\varphi(X) + b \tag{1}$$

where Y is the output vector, w is the weight vector, φ is the kernel function, X is the input vector, and b is the bias term [15].

3) Gradient Boosting Machine (GBM): Ensemble of decision trees, where each tree has the aim to correct the errors made by the previous and improve the final prediction. XGBoost (Extreme gradient boosting) and LightGBM (Light gradient boosting machine) models are newer version that improves the overall performance of the traditional GBM [14].

B. Neural Networks

1) Multilayer Perceptrons (MLP): Composed by layers of connected nodes (neurons) that transform inputs through weighted connections and non-linear activation functions. During the training, MLP adjust the weights to learn to approximate complex relationships between input features.

2) Recurrent Neural Network (RNN): A version of the standard artificial neural network (ANN) that process sequential data by maintaining a memory of previous inputs through internal states. RNNs are used for modeling time series data, since their ability to learn temporal dependencies.

3) Long Short-Term Memory (LSTM): Enhanced type of RNN designed to capture long-term dependencies in sequential data. LSTM consists of layers of memory cells, each cell contains three gates: forget, input, and output. The forget gate controls the information coming from the previous state that will no longer be stored, the input gate specifies the input features that will be stored in the next state, and the output gate specifies the information from the new state that will be output. This gated architecture enables LSTMs to effectively retain relevant information over long sequences, making them particularly suitable for modeling temporal patterns [3].

4) Convolutional Neural Network (CNN): Evolution of standard artificial neural network (ANN), CNNs are distinguished by their use of convolutional layers inspired by the mathematical operation of convolution. These layers filter the input by applying a mask of learnable weights to extract features and detect local patterns. In 1D CNNs, this mechanism can be applied to analyze time series data and extract meaningful features representing temporal variations. Moreover, 2D CNNs are widely used for image processing tasks due to their ability to identify spatial hierarchies and complex visual patterns [1].

IV. State of the Art

The review focused on the classification of recent research on the field of machine learning (ML) and deep learning (DL) usage for evapotranspiration (ET) estimation. With the goal of highlight the most commonly used approaches, 160 papers have been classified according to the model applied, performance metrics adopted, dataset used, and geographic areas covered.

A. Machine Learning and Deep Learning Models for ET Estimation

As shown in the figure 1, the classification reveals that Recurrent Neural Networks (RNN) models are the most frequently employed for ET estimation, since their strong capability of understanding temporal dependencies and seasonal patterns of the meteorological data that affects ET values.
This category includes several variants such as BiRNN, LSTM, BiLSTM, ConvLSTM, GRU, and BiGRU. Among these, the Long Short-Term Memory (LSTM) architecture is the most widely adopted, appearing in 75 studies (Table II).

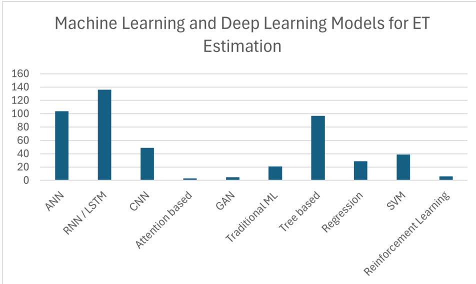


Fig. 1. Summary of algorithms used in the reviewed studies

TABLE II
RNN-BASED ALGORITHMS USED IN THE REVIEWED STUDIES

Algorithm	No. of Studies	Algorithm	No. of Studies
RNN	14	BiGRU	3
BiRNN	2	SSR-LSTM	3
LSTM	75	ConvLSTM	5
BiLSTM	18	NARX	1
GRU	15		

Other popular solutions are the standard Artificial Neural Networks (ANNs) (Table III), such as the Multi-Layer Perceptron (MLP) and Deep Belief Networks (DBN), as well as Tree based machine learning models like Random Forests (RF), M5Tree, and Gradient Boosting methods, including its optimized variants LightGBM and XGBoost (Table IV); see Figure 1. These models can be useful for understanding complex patterns in the climatological spatial data and their relationship with evapotranspiration ET values.

TABLE III
ANN-BASED ALGORITHMS USED IN THE REVIEWED STUDIES

Algorithm	No. of Studies	Algorithm	No. of Studies
DNN	52	DBN	7
MLP	6	DL	34
DeepFM	1	GRNN	2
BPNN	2		

TABLE IV
TREE BASED ALGORITHMS USED IN THE REVIEWED STUDIES

Algorithm	No. of Studies	Algorithm	No. of Studies
DT	5	GB	15
CART	1	XGB	14
BCART	1	LightGB	4
M5Tree	5	AdaBoost	1
RF	45	DRF	1
BT	1		

B. Performance Metrics Used

Performance metrics were classified considering how they evaluate the goodness of the predictions.
Figure 2 illustrates with an histogram the occurrences of each class of metrics in the reviewed papers. The most common class is the Squared Error metrics, which includes Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Root Mean Square Deviation (RMSD); see Table V.
These metrics penalize significant deviations between the predicted ET value and the real one..

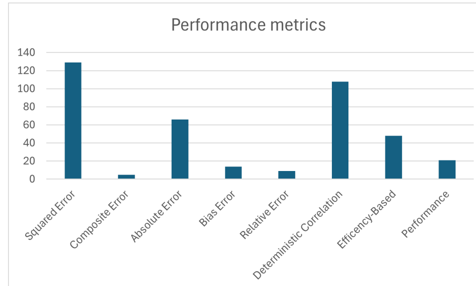


Fig. 2. Summary of performance metrics used in the reviewed studies

TABLE V
SQUARED ERROR METRICS USED IN THE REVIEWED STUDIES

Metric	No. of Studies
RMSE	100
MSE	28
RMSD	1

Another widely implemented class of metrics is the Deterministic Correlation, which refers to the correlation coefficient (R), the coefficient of determination (R^2), and Variance Accounted For (VAF); see Table VI.
These metrics are widely implemented to evaluate how well the predicted trends match the observed patterns.

TABLE VI
DETERMINISTIC CORRELATION METRICS USED IN THE REVIEWED STUDIES

Metric	No. of Studies
R	21
R^2	85
VAF	1

Additionally, several studies employed Absolute Error metrics, such as Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE), since their interpretability and robustness to outliers; see Table VII.

TABLE VII
EFFICENCY-BASED METRICS USED IN THE REVIEWED STUDIES

Algorithm	No. of Studies	Algorithm	No. of Studies
MAE	52	MADE	1
MAPE	9	MdAPE	3
WAPE	1	AARE	1
MAD	1		

Lastly, Efficiency-Based metrics, as the Nash–Sutcliffe Efficiency (NSE), the Kling–Gupta Efficiency (KGE), and the Index of Agreement (IOA), were also frequently used due to their ability to assess how well the model predicts relative to the mean of the observed data; see Table VIII.

TABLE VIII
EFFICENCY-BASED METRICS USED IN THE REVIEWED STUDIES

Algorithm	No. of Studies	Algorithm	No. of Studies
NSE	38	NNSE	1
KGE	6	IOA	3

C. Input Data Sources

ET prediction relies on a combination of meteorological data (e.g., temperature, humidity, solar radiation), ground measurements (e.g., water, energy, gas flux), and satellite derived surface indices (e.g., NDVI, EVI, LST).
Figure 3 shows that the primary source of data comes from ground based measurements and weather stations.
This source of data provide highly accurate and high frequency data, but their spatial coverage is limited to the locations where the measurements are taken, for this reason they are most useful in studies where the coverage area is local or regional (see figure 4 and figure 5).

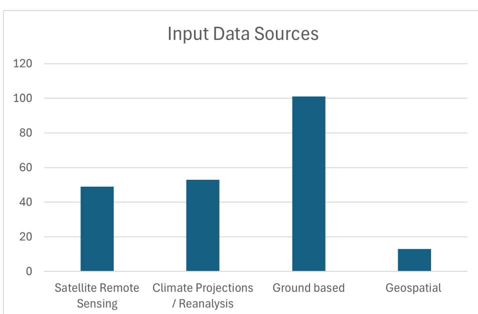


Fig. 3. Summary of data sources used in the reviewed studies

1) Ground based measurements: 101 papers over 160 have used ground based data to feed their prediction algorithms.
Between them, the most commonly used dataset is FLUXNET, a global network of micrometeorological tower sites distributed worldwide, where each uses the eddy covariance technique to measure exchanges of carbon dioxide, water vapor, and energy between terrestrial ecosystems and the atmosphere.
Weather stations collects meteorological data, mostly used to feed traditional reference methods (e.g. FAO-56 Penman-Monteith, Hargreaves-Samani) and provide standard ET estimates.

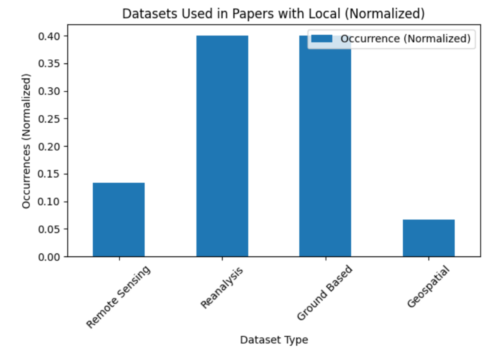


Fig. 4. Datasets Used in Papers with Local study areas

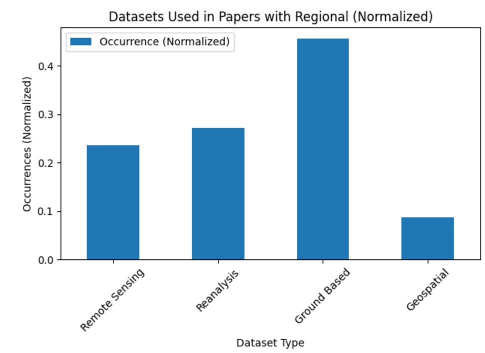


Fig. 5. Datasets Used in Papers with Regional study areas

In studies with a global study area, the most common choice is to use remote sensing based datasets or reanalysis datasets (see figure 6).

2) Satellite remote sensing datasets: Satellite products such as MODIS (Moderate Resolution Imaging Spectroradiometer), Copernicus, GRACE (Gravity Recovery and Climate Experiment), HyTES, and GLASS provide indices related to land surface temperature, vegetation coverage, and water storage changes. These datasets enable ET estimation over large areas but often at lower temporal or spatial resolutions.

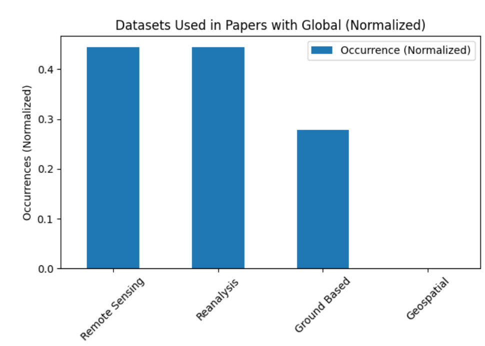


Fig. 6. Datasets Used in Papers with Global study areas

When the the study has a long-term projection, is important to rely on consistent, temporally extensive, and spatially comprehensive representations of atmospheric and surface conditions. For these studies the most convenient choice is to use climate model projections or reanalysis datasets (see figure 7).

3) Climate model projections and reanalysis datasets: Climate models, as CMIP6 (Coupled Model Intercomparison Project Phase 6), simulate future climate scenarios and provide projections of meteorological variables influencing ET.
Reanalysis datasets such as ERA5, GLDAS (Global Land Data Assimilation System), and CHIRPS (Climate Hazards Group InfraRed Precipitation with Station data) integrate satellite observations and model outputs to produce consistent, gridded climate and hydrological variables over extended periods.

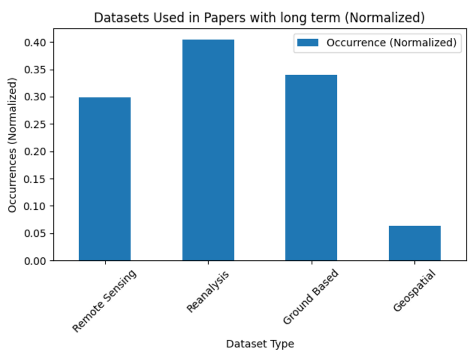


Fig. 7. Datasets Used in Papers with Long Term Projection

For local and regional studies, geospatial datasets could be a suitable choice as they provide information directly tied to the specific referred area.

4) Geospatial datasets: In this review only few studies have been found using these datasets. The reason is that, even if they are reliable and consistent, they are available only for some specific areas.
For example, some of these datasets found during the review are CAMELS (provides climate, land cover, soil, topography data for U.S. basins), CANOPEX (contains hydrological information about Canadian river basins), and TAMARACK (long-term hydrometeorological dataset used to support ecohydrological and snow process studies in cold regions).

D. Geographic and Temporal Distribution

The review highlights that the majority of evapotranspiration studies are located in Aisa (see figure 8).
Particularly, the Asian countries that appeared the most are China and India, due to their vast agricultural regions stressed by the high demand and by the different climatological conditions.
Outside the Asiatic continent, other popular countries found during the review are the United States and Australia. Both cover a wide geographic area characterized by significant climatic variability and high agricultural water requirements.
Only few studies have been found focusing over European and Mediterranean regions. The reason is that in these regions is difficult to perform ET prediction due to several environmental factors, as the intense seasonal droughts, fragmented land cover, and heterogeneous climatic patterns.

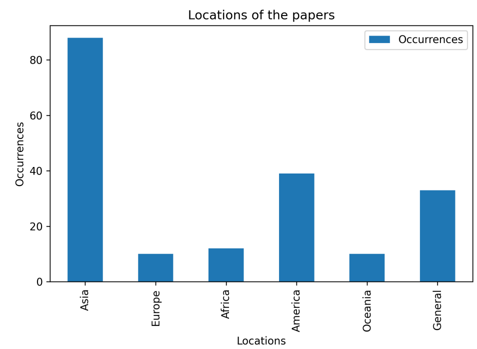


Fig. 8. Summary of locations of the studies

E. Preprocessing and Feature Engineering Techniques

A common issue in ET prediction datasets used to train a neural network, is the presence of relationships among the input variables, that can cause the model to overfit and have bad prediction performances. These datasets contains directly related features as meteorological values, vegetation indices and surface attributes.
Therefore, it is common to reduce collinearity, presence of linear relationships among input variables, using preprocessing algorithms such as Principal Component Analysis (PCA). Independent Component Analysis (ICA) may also be used to extract statistically independent components.
Other relevant algorithms found are the PSO (Particles Swarm Optimization), used to find the best hyperparameters for the neural network, and SHAP (SHapley Additive exPlanations) that interprets model predictions and the importance of each feature.

V. Materials and Methods

After the review, a personal implementation of a neural network has been developed and tested to perform ET predictions.
The goal was to use all the knowledge obtained during the review to implement a reliable and efficient model to predict the actual evapotranspiration (AET) over the Piedmont region in Italy.

A. Dataset and Study Area

The development of the neural network started by finding the best dataset to perform AET estimations in Italy. As the review shows, few studies have implemented evapotranspiration model in Italy due to the climatological complexity of the mediterranean area.
Moreover, evapotranspiration (ET) trend in the Piedmont region, located in the northern Italy, is characterized by high

spatial and temporal variability influenced by its position between Mediterranean warm areas and more temperate alpine zones [16] [17]. To address this challenge, two preprocessed datasets have been used, the first contains climatological variables used to perform regression, and the second for the target AET values. These two dataset have been aligned in space and time, to then obtain an unique dataset that for each pixel (longitude, latitude) of Italy contains 8 climatological features (minimum, average, and maximum air temperature, minimum and maximum relative humidity, wind speed, surface solar radiation, precipitation) and the related AET value for each day. The study focused only on the positions of the Piedmont region and in a time period that elapse from 01-01-2018 to 01-01-2022.

1) MADIA - Meteorological variables for agriculture: A dataset for the Italian area: This dataset provides an high-resolution series of agro-meteorological variables across Italy, gridded at a 0.25-degree resolution, and derived from ERA5 reanalysis data for the period 1981–2022.
It contains daily time series of minimum, average, and maximum air temperature, minimum and maximum relative humidity, wind speed, surface solar radiation, precipitation, and the reference evapotranspiration (ET_0) calculated using the FAO Penman-Monteith method.
The MADIA dataset is owned and maintained by the Council for Agricultural Research and Economics (CREA), an Italian public research institution focused on agriculture and environment.
The dataset is described and available in the paper [18].

2) GLEAM (Global Land Evaporation Amsterdam Model): GLEAM is a satellite-based modeling framework that provides global, daily estimates of terrestrial evapotranspiration (ET) and its components at 0.25° resolution.
It uses the Priestley–Taylor equation (or Penman in newer versions) to calculate the potential evapotranspiration (PET), which estimates the maximum possible ET assuming unlimited water availability.
It adjusts PET with an evaporative stress factor based on vegetation optical depth (VOD), an index for vegetation water content, and a multi-layer soil moisture model. GLEAM outputs actual evapotranspiration AET by summing evaporation and transpiration components, with data available at https://www.gleam.eu/ (accessed May 25, 2025), where registered users can download data via SFTP.

3) Datasets Alignment: MADIA dataset was downloaded as a zip folder from https://zenodo.org/records/7252361 (accessed May 27, 2025), as referenced in the related publication [18]. The folder contains a CSV file for each year from 1981 to 2022, and each file provides daily records for multiple spatial grid points across Italy, with 8 meteorological features and auxiliary information for each date and location (see Table IX).

Since this study focuses exclusively on the Piedmont region, a spatial filter was applied to select only the grid points located within the area bounded by latitudes 44.0° to 46.5° and longitudes 6.5° to 9.0°.

TABLE IX
MADIA DATASET DESCRIPTION

Acronym	Description
longitude	longitude coordinate
latitude	latitude coordinate
time	date of the day
tasmin	mean of daily minimum near-surface air temperature
tasmax	mean of daily maximum near-surface air temperature
tasmean	mean of daily average near-surface air temperature
rhmin	mean of daily minimum near-surface relative air humidity
rhmax	mean of daily maximum near-surface relative air humidity
ws10	mean of daily wind speed
ssrd	mean of daily surface solar radiation (shortwave radiation)
ppn	sum of daily water precipitation
pev	sum of daily crop ET_0 estimated by FAO Penman-Monteith
expver	code which identifies temporary data when expver=5
mask	boolean code to identify cells belonging to the Italian country

Figure 9 displays all the spatial points available in the dataset, highlighting in red those corresponding to the Piedmont region. Additionally, the dataset was temporally filtered to include data only from 2018 to 2022.

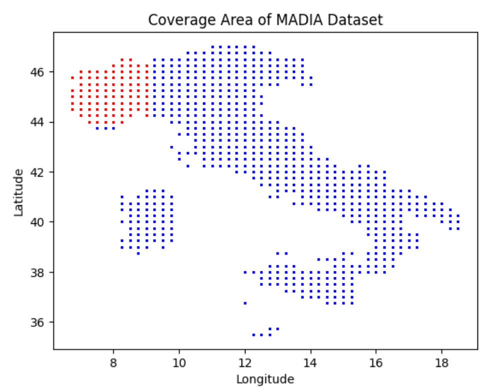


Fig. 9. Spatial Points of the filtered MADIA Dataset

GLEAM dataset was obtained via an SFTP connection, through which annual NetCDF (NC) files can be downloaded, each containing daily values for a specific variable. For this study, the NC files corresponding to daily AET values from 2018 to 2022 were downloaded.
However, a challenge emerged due to the global spatial coverage of the dataset. As a result, each NC file was heavy to be downloaded and handled, so the same spatial filter used for MADIA dataset was applied to extract only the grid points related to Piedmont.
The final dataset consists of 1,826 days (five years, including one leap year) and a spatial grid of 25 latitude points by 22 longitude points. This results in 550 pixels per day (25×22). Figure 10 shows the AET values contained in the extracted grid points over the Piedmont region in the first day of

the dataset. Red pixels (14 in total) highlights Null values corresponding to the sea.

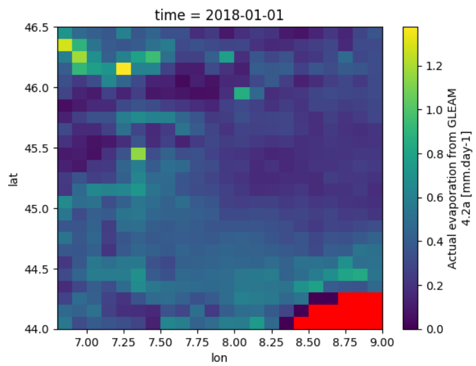


Fig. 10. Spatial Points of the Filtered GLEAM Dataset

Although the two datasets refer to the same geographic area, their spatial points are not perfectly aligned. To construct an unified dataset, each pixel in the MADIA dataset was matched, for each day, with the AET value of the nearest spatial point in the GLEAM dataset. As a result, the MADIA dataset has a new feature representing the daily AET value for each pixel. Table X summerized the minimum, average, and maximum spatial shifts applied during the alignment.

TABLE X
MINIMUM, MEAN, AND MAXIMUM SPATIAL SHIFT (IN DEGREES) APPLIED DURING DATASET ALIGNMENT

Distance	Degrees
Min.	0.0
Max.	0.112
Mean	0.049

Figure 11 illustrates the final dataset, composed by the same grid points of the MADIA dataset but with the associated AET value.

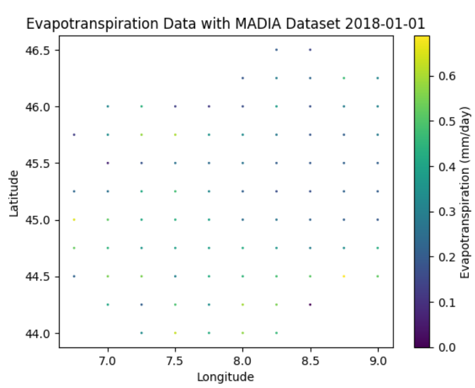


Fig. 11. Spatial Points of the Final Dataset

B. Data Preprocessing

Before training the neural network, several preprocessing steps were applied to enhance data quality, reduce redundancy, and address missing values.

1) Handling Missing Values: The dataset may contain missing (Null) values in the AET field, particularly in pixels located near coastal areas or where GLEAM does not provide data. To ensure consistent input across the entire study area, all records with missing AET values were removed from the dataset (see Table XI).

TABLE XI
SAMPLE COUNT BEFORE AND AFTER REMOVING MISSING *AET* VALUES

Dataset Status	Number of Samples
Before Removing Missing Values	160,688
After Removing Missing Values	158,862

2) Normalization: To prepare a dataset to feed a neural network, normalization is a crucial step to ensure equal contribution of each feature during the learning process, and improves convergence time during training.
In this study, both the input features and the target variable (AET) were normalized using the Standard Scaler. This method transforms each feature to have zero mean and unit variance according to the formula 2:

$$x' = \frac{x - \mu}{\sigma} \tag{2}$$

where:

- x is the original feature value,
- μ is the mean of the feature,
- σ is the standard deviation of the feature.

The final set of input features and the target variable used in the regression task is listed in Table XII.

TABLE XII
FINAL DATASET STRUCTURE

Feature	Description
day_of_year	Integer (1–365/366)
tasmin	Daily minimum temperature [°C]
tasmean	Daily mean temperature [°C]
tasmax	Daily maximum temperature [°C]
rhmin	Daily minimum relative humidity [%]
rhmax	Daily maximum relative humidity [%]
ws10	Wind speed at 10 meters [m/s]
ssrd	Surface solar radiation [W/m²]
ppn	Precipitation [mm/day]
E	Actual Evapotranspiration [mm/day] (target variable)

3) Feature Correlation Analysis: A first experiment has revealed an unexpected behavior during the training phase of the neural network, the loss function (evaluated using both Mean Squared Error and Huber loss) dropped to very small values after just a single epoch, indicating potential overfitting or a simplistic learning scenario.
To investigate the cause, a correlation analysis was performed using a correlation matrix (see Figure 12). This matrix measures the linear relationships between pairs of features, where values close to ±1 indicate strong positive or negative correlations that can cause the network to redundantly learn the same patterns and so overfit.

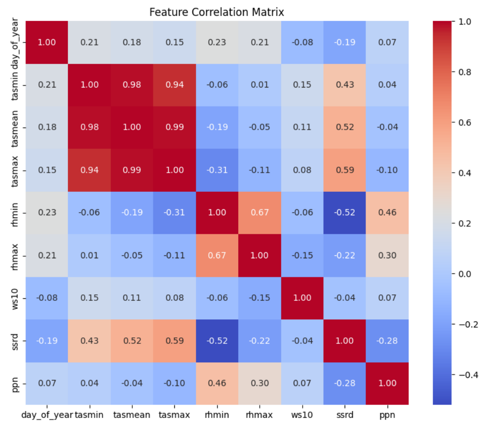


Fig. 12. Features Correlation Matrix

To address this issue, Principal Component Analysis (PCA) algorithm has been used to transforms the original set of correlated features into a new set of uncorrelated components called principal components (see Table XIII).

TABLE XIII
SAMPLE COUNT BEFORE AND AFTER REMOVING MISSING *AET* VALUES

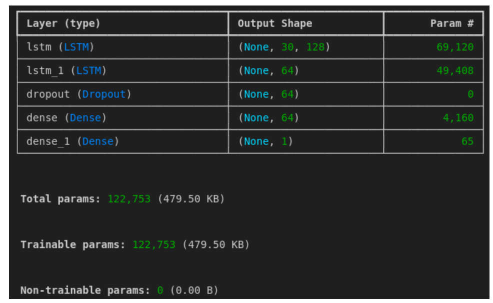
Dataset Status	Number of Features
Before PCA	9
After PCA	6

C. Neural Network Architecture

By training exclusively on temporal trends, the model learns to predict the AET of the following day based only on the climatological evolution of the previous days, without considering the location and the data of the next day.
To perform this task, a Long Short-Term Memory (LSTM) network was developed, a type of Recurrent Neural Network (RNN) particularly effective in modeling temporal dependencies and sequential patterns. The LSTM architecture was developed in Python by using the `TensorFlow` library, and the source code is publicly available at https://github.com/CavaWasTaken/InnovativeWirelessPlatformsForIoT.
The final network is composed by five layers, totaling 122,753 trainable parameters:

- **First Layer:** An LSTM layer that receives a batch of sequences, each consisting of 30 days of climatological data (sequence length defined as a hyperparameter). The features of each day in the sequence is processed by an LSTM cell, and each cell (except the first) also receives input from the previous cell to model temporal dependencies. This layer uses 128 units and is configured with `return_sequences=True`, meaning that its output will be composed by the outputs of each cell, and its shape is $(\text{batch\_size}, 30, 128)$.
- **Second Layer:** A second LSTM layer that processes the sequence output from the first layer. This layer uses 64 units and has `return_sequences=False`, so it returns only the output of the final cell, with shape $(\text{batch\_size}, 64)$.
- **Third Layer:** A Dropout layer that randomly disables 10% of the neurons during training to help prevent overfitting.
- **Fourth Layer:** A Dense (fully connected) layer with 64 units and ReLU activation, which transforms the input from the previous layer into a higher-level representation.
- **Fifth Layer:** A final Dense layer with 1 unit, which performs the regression task and outputs the predicted AET value.

The full architecture is illustrated in Figure 13.



Layer (type)	Output Shape	Param #
lstm (LSTM)	(None, 30, 128)	69,120
lstm_1 (LSTM)	(None, 64)	49,408
dropout (Dropout)	(None, 64)	0
dense (Dense)	(None, 64)	4,160
dense_1 (Dense)	(None, 1)	65

Total params: 122,753 (479.50 KB)

Trainable params: 122,753 (479.50 KB)

Non-trainable params: 0 (0.00 B)

Fig. 13. LSTM Architecture

VI. EXPERIMENTS AND EVALUATION

A. Sequence Construction

Since LSTM model requires to receive as input sequences of data, the dataset was segmented into sequences of 30 consecutive days of daily climatological variables, and each sequence was paired with the AET value of the following day.
The sequences were randomly shuffled to prevent the model from overfitting seasonal trends. An 80%–20% train-test split was applied to the sequences (see Table XIV).

TABLE XIV
TRAIN AND TEST SET SPLITTING

Set	Features Shape	Target Variable Shape
Original	(156,252; 30; 6)	(156,252;)
Train	(125,001; 30; 6)	(125,001;)
Test	(31,251; 30; 6)	(31,251;)

B. Training Phase

The model was trained over 50 epochs using a batch size of 64 sequences. A validation test of 20% of the training data was used to monitor the model's performance during the training phase.
The optimization was performed using the Adam optimizer and the Mean Squared Error (MSE) was used as the loss function. Each epoch involved 1563 iterations and required approximately 18 seconds, totaling 932.28 seconds for the full training phase. The complexity of the model required the usage of Google Colab GPU capabilities to perform the training.

C. Testing Phase and Results

Testing was performed on the remaining 20% of the dataset. Since the model was trained on normalized data, the predicted AET values needs to be rescaled to their original range for better understanding.
Figure 14 shows a scatter plot comparing the predicted AET values to the true values. The red line indicates the ideal 1:1 regression line, helping visualize the goodness of the test.

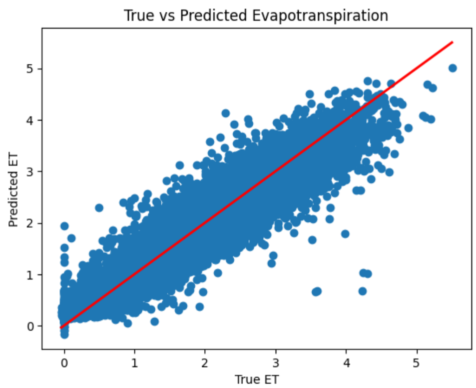


Fig. 14. LSTM Regression Predictions vs. Ground Truth

Three metrics were used to assess prediction performance:

- R^2 **(Coefficient of Determination)** – Can take values from 0 to 1, closer is to 1 and more the predicted values replicate the trend of the ground truth values.

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \quad (3)$$

- **RMSE (Root Mean Squared Error)** – Evaluates the amount of error in the same units as the target variable.

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \quad (4)$$

- **MAE (Mean Absolute Error)** – Represents the average absolute difference between predicted and true values.

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \quad (5)$$

The values obtained from the evaluation metrics offer a clear view of the model's performance. R^2 scores a value of 0.9392, indicating that nearly 94% of the variance in the actual AET values is explained by the model's predictions. This confirms that the model has learned to capture the underlying temporal patterns effectively.
The RMSE of 0.2697 [mm/day] and MAE of 0.2000 [mm/day] indicate consistent and relatively small prediction errors across the dataset.
The results are summarized in Table XV.

TABLE XV
EVALUATION METRICS ON TEST SET

Metric	Value
R^2	0.9392
RMSE	0.2697
MAE (%)	0.2000

D. Other Tests

Additional tests were performed to understand the capabilities of the model and explore simpler approaches for reliable AET prediction.
The primary model employed Principal Component Analysis (PCA) to reduce dimensionality and mitigate multicollinearity, but depends on the availability of the full set of climatological variables. For this reason, an additional model has been implemented to test if is possible to perform AET estimation without all the meteorological variables.
To do so, the correlation matrix (see Figure 12) was used to detect and drop those features having a correlation coefficient that exceeds a defined threshold of ± 0.6. In this case mean and maximum air temperature, and maximum relative humidity were excluded, resulting in a final set made of 6 input features. During the testing phase, the simplified model achieved performance comparable to the PCA-based model. The regression plot for this model is presented in Figure 15. Additionally, the simplified model was evaluated using shorter input sequences consisting of 14 consecutive days of meteorological data, and still demonstrates promising results, with a slight drop in the performances.
Finally, a transferability test was conducted to assess whether a model trained over the Piedmont region could generalize to a different geographic area. This test was conducted over the Venetian region, located in the north-east of Italy, and results in significantly lower performance, suggesting that the model strongly learns temporal patterns of the region.
All the metric values obtained for each test are shown in the table XVI.

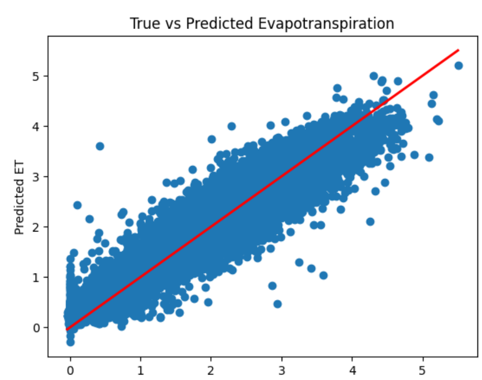


Fig. 15. LSTM Regression Predictions vs. Ground Truth - Reduced Dataset Test

TABLE XVI
EVALUATION METRICS ON TEST SET FOR ALL MODEL CONFIGURATIONS

Test Configuration	R^2	**RMSE**	**MAE**
PCA-based model (30 days)	0.9392	0.2697	0.2000
Feature-selected model (30 days)	0.9414	0.2648	0.1940
Feature-selected model (14 days)	0.9140	0.3213	0.2285
Transferability to Venetian region (30 days)	0.5744	0.7125	0.4625

VII. CONCLUSIONS

This study proposes a temporal based deep learning approach for predicting daily Actual Evapotranspiration (AET) using historical sequences of climatological variables. The developed Long Short-Term Memory (LSTM) model demonstrates the ability to predict next-day AET based on 30 consecutive days of meteorological data. The proposed model proved to be a reliable solution that can work even without some climatological features, thus reducing the need for dense sensor networks or complete datasets.

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